

Low-cost TMS: Transcranial Magnetic Stimulation for Less Than \$1000

Article title

Low-cost TMS: Transcranial Magnetic Stimulation for Less Than \$1000

Authors

Maxence Lapatrie^{1,2,*}, Antonella Catanzaro^{3,4}, Yuzuha Isetani⁴, Andrew Lyu⁵, Han C. Nguyen¹, and Jathav Puvirajan⁶

Affiliations

¹Department of Electrical and Computer Engineering, McGill University, Montreal, Canada

²McConnell Brain Imaging Centre, Montreal Neurological Institute, Montreal, Canada

³Department of Physiology, McGill University, Montreal, Canada

⁴Department of Computer Science, McGill University, Montreal, Canada

⁵TODO

⁶TODO

Corresponding author's email address

maxence.lapatrie@mail.mcgill.ca

Abstract

Max. 200 words. Remember that the abstract is what readers see first in electronic abstracting & indexing services - make it brief, specific, interesting and easy to understand. If a research article refers to your hardware, cite that research article here.

Keywords

brain stimulation, low-cost, open source, diy, transcranial magnetic stimulation, TMS

Specifications table

Please replace the italicized instructions in the right column of the table with the relevant information about your hardware.

Hardware name	Monophasic Transcranial Magnetic Stimulator
Subject area	• Engineering and material science • Neuroscience
Hardware type	• Imaging tools • Electrical engineering and computer science
Closest commercial analog	Magstim M200²
Open source license	CERN-OHL-P
Cost of hardware	\$601.95 (USD)
Source file repository	<i>If you've uploaded your source files to an approved repository (OSF, Mendeley Data or Zenodo) write the DOI URL here. For exemple: http://doi.org/10.17605/OSF.IO/WGK7Q</i>

1. Hardware in context

Write a short description of the hardware and provide context, i.e., describe similar open hardware and proprietary equipment in the field.

2. Hardware description

Describe your hardware, highlighting the customization rather than the steps involved in the procedure. Explain how it differs from other hardware and the advantages it offers over pre-existing methods. For example, how does this hardware compare to other hardware in terms of cost or ease of use, or how can it be used to develop further designs in a particular area?

Add 3-5 bullet points which broadly explain to other researchers - inside or outside of the original user community - how the hardware could help them, with either standard or novel laboratory tasks.

Design files

Your design files should be editable - see [OSHOWA's open source definition of 'Documentation'](#) for further details. You must then either:

- Upload your design files to one of the three approved online repositories - [Mendeley Data \(instructions\)](#), the [Open Science Framework \(instructions\)](#) or [Zenodo \(instructions\)](#). We recommend this option as the repositories support versioning of files.*
- Upload your design files as supplementary materials (e.g., CAD files, videos...) to Hardware X's online editorial system when you submit your manuscript.*
- Include your design files in the body of the manuscript (e.g., as figures).*

CAD files: You are encouraged to use free and open source software packages for creating the files. For CAD files, [OpenSCAD](#), [FreeCAD](#), or [Blender](#) are encouraged, but, if these are not available, we accept source files from proprietary CAD packages, such as Autocad or Solidworks, and other drawing packages.

3D printing: Supplementary files that facilitate digital replication of the devices are encouraged; for example, STL files for 3D printing components. We recommend uploading CAD files to the [NIH 3D Print Exchange](#) as Custom Labware and then entering the link [here](#).

Electronics: PCB layouts and other electronics design files can be uploaded to the [Open Hardware Repository](#) or other repositories or as supplementary materials.

Software and firmware: All software files used in the design and operation of the hardware should be included in the repository. Provide a description of the software and firmware and use extensive comments in the code.

3. Design files summary

Complete a separate row for each design file associated with your hardware (including the primary design files). Any empty rows should be deleted.

Design filename	File type	Open source license	Location of the file
discharge.kicad_sch	KiCAD schematic file	CERN-OHL-P	discharge.kicad_sch
charge.kicad_sch	KiCAD schematic file	CERN-OHL-P	charge.kicad_sch
gate_driver.kicad_sch	KiCAD schematic file	CERN-OHL-P	gate_driver.kicad_sch
tms_schematic.pdf	PDF file	CERN-OHL-P	tms_schematic.pdf
discharge.asc	ASCII schematic file	CERN-OHL-P	discharge.asc
charge.asc	ASCII schematic file	CERN-OHL-P	charge.asc
gate_driver.asc	ASCII schematic file	CERN-OHL-P	gate_driver.asc
linear_optocoupler.asc	ASCII schematic file	CERN-OHL-P	linear_optocoupler.asc
TMS_holder.FCStd	FreeCAD parametric CAD file	CERN-OHL-P	TMS_holder.FCStd
sensing_cross.FCStd	FreeCAD parametric CAD file	CERN-OHL-P	sensing_cross.FCStd
coil_winder.FCStd	FreeCAD parametric CAD file	CERN-OHL-P	coil_winder.FCStd
TMS_field_grid.pdf	PDF file	CERN-OHL-P	TMS_field_grid.pdf
controller_arduino.ino	Arduino sketch file	MIT	controller_arduino.ino
app.py	Python script file	MIT	app.py
collect_grid.py	Python script file	MIT	collect_grid.py
data_analysis.ipynb	Jupyter notebook file	MIT	data_analysis.ipynb

KiCAD schematic files

discharge.kicad_sch - Schematic overview of the entire TMS circuit with focus on the TMS discharge circuit.

charge.kicad_sch - Main charging circuit schematic with measurements and control sub-circuits.

gate_driver.kicad_sch - Gate driver sub-circuit schematic.

tms_schematic.pdf - Entire TMS circuit schematic in PDF format.

LTSpice schematic files

discharge.asc - TMS inductor-capacitor oscillator discharge circuit schematic for SPICE simulation.

charge.asc - ZVS capacitor charging circuit schematic for SPICE simulation.

gate_driver.asc - Gate driver circuit schematic for SPICE simulation.

linear_optocoupler.asc - Linear optocoupler measurement circuit schematic for SPICE simulation.

FreeCAD parametric CAD files

TMS_holder.FCStd - Elevating table to hold the TMS coil for field measurements.

sensing_cross.FCStd - Sensing cross to translate user movements to pick up coil movements.

coil_winder.FCStd - Coil winder for winding the TMS coil.

TMS_field_grid.pdf - Measurement grid for optimal pick up coil positioning.

Control software files

controller_arduino.ino - Arduino sketch for controlling the TMS circuit.

app.py - Python script for the user interface.

Data collection and analysis files

collect_grid.py - Software interface with the oscilloscope for collecting pick up coil measurements.

data_analysis.ipynb - Jupyter notebook for analyzing the data acquired with the TMS.

Bill of materials

If your bill of materials is long or complex, you can upload the details in an editable spreadsheet, e.g., ODS file type, Excel spreadsheet or PDF file, to an open access online location, such as the [Open Science Framework](#) repository. Include the link here. Alternatively, the bill of materials can be submitted alongside your manuscript as supplementary material.

4. Bill of materials summary

Designator	Component	Number	Cost per unit - currency	Total cost - currency	Source of materials	Material type
If possible, use the same designator here as you use in the associated design file. If that's not possible, you will need to explain the relationship between the two.	Name of Component 1	Number of units	Cost per unit and the currency used	Total cost and the currency used	If possible, include a direct link to a webpage where the component can be purchased	Select from: -Metal -Semi-conductor -Ceramic -Polymer -Biomaterial -Organic -Inorganic -Composite -Nanomaterial -Semiconductor -Non-specific -Other
...	...	...	...	...	...	...
...	...	...	...	...	...	...
...	...	...	...	...	...	...

You can use this space for any additional descriptions of the materials used.

5. Build instructions

Provide detailed, step-by-step construction instructions for the submitted hardware:

- Include all necessary information for reproducing it.
- Explain and (when possible) characterize design decisions. Include any design alternatives you created.
- Use visual instructions such as schematics, images and videos.

- Clearly reference design files and component parts described in the **Design file summary** and the **Bill of materials summary**.
- Highlight any potential safety concerns.

6. Operation instructions

Provide detailed, step-by-step instructions for the safe and proper operation of the hardware.

- Use visual instructions, as necessary.
- Highlight any potential safety hazards.

7. Validation and characterization

Demonstrate the operation of the hardware and characterize its performance for a specific scientific application.

- Highlight a relevant use case.
- If possible, characterize performance of the hardware over operational parameters.
- Create a bulleted list describing the capabilities (and limitations) of the hardware. For example, load and operation time, spin speed, coefficient of variation, accuracy, precision, etc.

Ethics statements

HardwareX has ethical guidelines that all authors must comply with. In addition, we ask you to complete the relevant statement(s) below. Please delete those which are not relevant for your work.

If your work involved human subjects, please include a statement here confirming that the relevant informed consent was obtained from those subjects:

If your work involved animal experiments, please include a statement here confirming that those experiments complied with the ARRIVE guidelines and were carried out in accordance with the U.K. Animals (Scientific Procedures) Act, 1986 and associated guidelines; EU Directive 2010/63/EU for animal experiments; or the National Institutes of Health guide for the care and use of laboratory animals (NIH Publications No. 8023, revised 1978). Note, the sex of the animals must be indicated, and, where appropriate, the influence (or association) of sex on the results of the study:

CRediT author statement

CRediT is an initiative that enables authors to share an accurate and detailed description of their diverse contributions to a published work.

Example of a CRediT author statement: **Zhang San**: Conceptualization, Methodology, Software **Priya Singh**: Data curation, Writing- Original draft preparation. **Wang Wu**: Visualization, Investigation. **Jan Jansen**: Supervision. **Ajay Kumar**: Software, Validation.: **Sun Qi**: Writing- Reviewing and Editing.

Please add a CRediT author statement for your data article here, using the [categories listed on this webpage](#).

Acknowledgements

All contributors who do not meet the criteria for authorship should be listed in an acknowledgments section.

In addition, please list any funding sources in this section. List funding sources in this standard way to facilitate compliance to funder's requirements:

Funding: This work was supported by the National Institutes of Health [grant numbers xxxx, yyyy]; the Bill & Melinda Gates Foundation, Seattle, WA [grant number zzzz]; and the United States Institutes of Peace [grant number aaaa].

It is not necessary to include detailed descriptions on the program or type of grants and awards. When funding is from a block grant or other resources available to a university, college, or other research institution, submit the name of the institute or organization that provided the funding.

If no funding has been provided for the research, please include the following sentence:

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

References

If relevant, you should include a reference to the original publication of the hardware you customized and a reference to the repository in which your design files are published. Other references can be included, as required; for example, references that put your device in context in the literature. For more information on the reference format in HardwareX please see the [Guide for Authors](#).

Additional Information for authors. (do not include these lines in your submission)

Author manuscript checklist

- *Is the subject of the submission under an open source license? Are design files in the preferred format for making modifications as defined by the [Open Source Hardware definition](#)?*
- *Can the hardware be reproduced with the details provided in the submission?*
- *Are all relevant design files available on either the Mendeley Data, Open Science Framework, or Zenodo repository? Are they described in the Design Files Summary, and clearly documented? (E.g., descriptive file names, commented code, labeled images, etc.)*
 - *If in the Open Science Framework, has the repository been registered? [Instructions](#)*
 - *If in Zenodo, is the repository open access and published? [Instructions](#)*
 - *If in Mendeley Data, is the repository published or the sharable link included in the additional information you plan to submit? [Instructions](#)*
- *Are visual instructions used when necessary?*
- *Is the utility of the hardware to the scientific community explained clearly? Has a specific scientific application been demonstrated using the hardware?*
- *Is the performance of the hardware adequately demonstrated and characterized?*
- *Are all potential safety concerns addressed?*
- *For more information on the article template consult the [Guide to Authors](#).*

Reminder: Before you submit, please delete all the instructions in this document (the text in italics), including this paragraph.

Thank you!